Paper 1

Mathematics 4052/01
October/November 2023
2 hours 15 minutes

Answer **all** *the questions.*

1. Sima and Ken share \$480 in the ratio 9 : 7.
   Find the amount each receives.

Answer Sima \$ ....
Ken \$ .... [2]

2. (a) $\sin x° = 0.9301$

   Find two possible values of x in the range $0 \leqslant x \leqslant 180$.

Answer .... or .... [2]

   (b) Convert 1.36 radians into degrees.

Answer .... [1]

3. (a) Simplify $(2c^3d^2)^3$.

Answer .... [1]

   (b) $16 \times 5^4 + 9 \times 25^2 = 5^k$

   Use the laws of indices to find the value of k.
   Show your working.

Answer k = ... [2]

4. (a) $M = p^{16} \times q^{8} \times r^{12}$ where p, q and r are prime numbers.

Explain why M is a perfect square.

....

.... [1]

(b) The number $S = 2^{8} \times 7^{10} \times 11^{15}$.

(i) The number $N = S \times k$ is a perfect cube.

Find the smallest possible integer value of k.

Give your answer as a product of its prime factors.

Answer $k =$ .... [1]

(ii) The number $T = 2^{20} \times 7^{8} \times 11^{15}$.

Find the lowest common multiple (LCM) of S and T as a product of its prime factors.

Answer .... [1]

5. Ravi invests \$4500 at a rate of 0.25% per month compound interest.

(a) Explain why the first year's interest is not 3% of \$4500.

....

.... [1]

(b) Calculate the value of his investment at the end of 12 months.

Answer \$ .... [2]

6. The expression $x^2 - 14x + b$ is equivalent to $(x + a)^2 - 25$.

(a) Find the value of a and the value of b.

Answer a = ...

b = ... [2]

(b) The curve $y = x^2 - 14x + b$ is drawn.

Write down the equation of the line of symmetry of the curve.

Answer .... [1]

7. **(a)** Adrian has 550 stamps in his collection.

He increases his number of stamps by 18%.

Find the number of stamps in his collection now.

Answer .... [2]

(b) Mai collects postcards.

28% of the postcards are from Australia.

55% of the remaining postcards are from Europe.

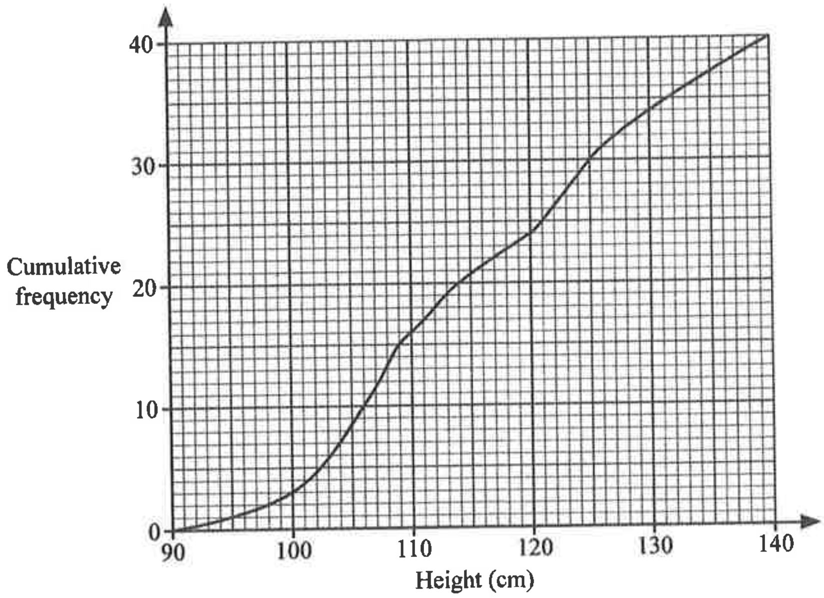
The other 81 postcards are from Asia.

Calculate the total number of postcards in her collection.

Answer .... [3]

8. A group of 40 children visit a theme park.

The cumulative frequency diagram represents their heights.



(a) Use the diagram to estimate

(i) the median height of the children,

Answer .... cm [1]

(ii) the interquartile range of the heights.

Answer .... cm [2]

(b) To be allowed on a particular ride at the theme park, children must be of a minimum height, h cm.

Only 15 of these children are allowed on the ride.

Find the value of h.

Answer h = ... [1]

(c) The information on the cumulative frequency diagram is shown in this table.

Height (H cm)	$90 < H \leqslant 100$	$100 < H \leqslant 110$	$110 < H \leqslant 120$	$120 < H \leqslant 130$	$130 < H \leqslant 140$
Frequency	3	13	8	10	6

(i) Calculate an estimate of the mean height of the children.

Answer .... cm [1]

(ii) Calculate an estimate of the standard deviation of the heights.

Answer .... cm [1]

9. Anhaa travels from Singapore to France.

She wants to change 780 Singapore dollars (\$) to euros (€).

She would receive €1 more if she changes the money in Singapore.

The exchange rate in France is €1 = \$1.5953.

Calculate the exchange rate in Singapore.

Give your answer in euros per Singapore dollar.

Answer \$1 = €........................ [2]

10. A is the point $(-2, 3)$ and B is the point (p, q).

The gradient of the line AB is $\frac{1}{3}$.

Find p in terms of q.

Give your answer in its simplest form.

Answer p = ... [2]

11. Write as a single fraction in its simplest form $\frac{3a}{4a-1} - \frac{2}{3a+1}$.

Answer .... [2]

12. **(a)** Factorise completely.

(i) $x^3y + xy^3$

Answer .... [1]

(ii) $15cd - 10ce + 12d^2 - 8de$

Answer .... [2]

(b) Expand and simplify $(2x + 3a)(5x - 2a)$.

Answer .... [2]

13. A bag contains 12 red marbles, 7 white marbles and 6 blue marbles.

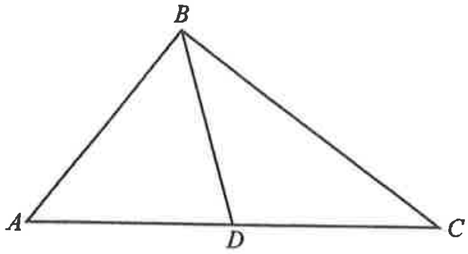
A further n white marbles are added to the bag.

The probability of picking a white marble is now $\frac{3}{5}$.

Find the value of n.

Answer $n =$ .... [2]

14.



ABC is a triangle.

D is a point on AC such that it is equidistant from A, B and C.

Explain why ABC is a right angle.

....

....

....

....

....

.... [2]

15. Explain why $(2n-1)^2 - 2(n-5)(2n-1)$ is a multiple of 3 for all integer values of n.

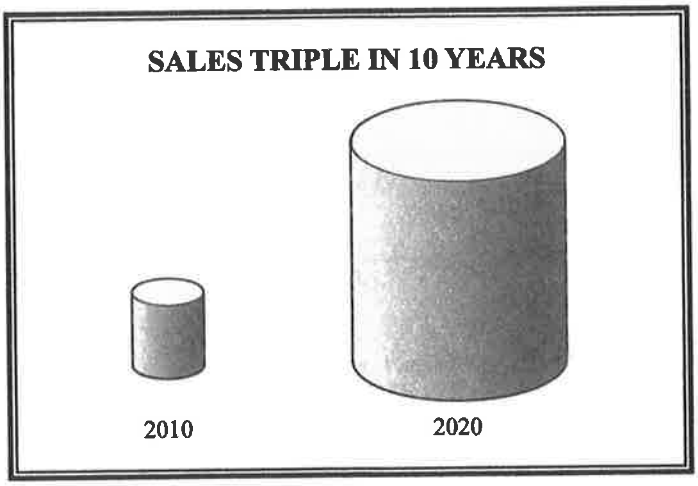
Answer

....

.... [3]

16. A company sells tinned food.

The company displays the sales figures with this chart.



The chart shows drawings of two geometrically similar solids. The dimensions of the tin for 2020 are three times the corresponding dimensions of the tin for 2010.

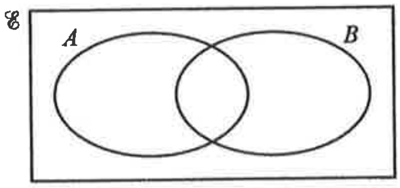
The chart is misleading.

Explain what it implies about the company's sales of tinned food.

....

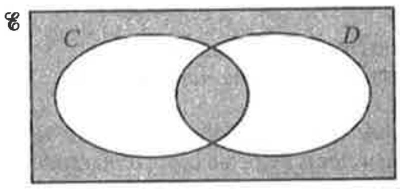
.... [1]

17. **(a)** On the Venn diagram, shade the region which represents $A \cup B'$.



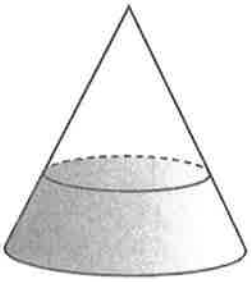
[1]

(b) Use set notation to describe the shaded region.



Answer .... [1]

18.



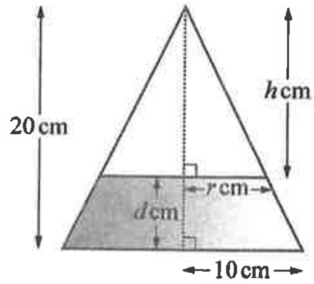


Diagram A shows a sealed cone containing some water.

Diagram B shows the cross-section of the cone and water.

The radius of the base of the cone is 10 cm and the height of the cone is 20 cm.

The radius of the surface of the water is r cm, the depth of the water is d cm and the surface of the water is h cm below the top of the cone.

(a) Show that the depth of the water, d, is $20 - 2r$.

Answer

[1]

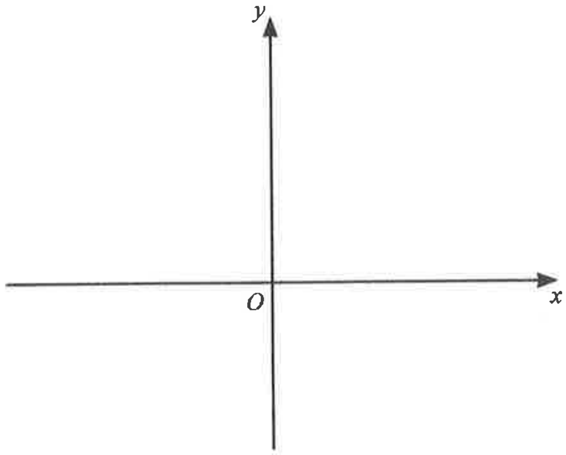
(b) The volume of water in the cone is equal to the volume of the empty space in the cone.

Calculate d.

Answer $d =$ .... [3]

19. Sketch the graph of $y = -(x - 3)(x + 7)$ on the axes below.

Indicate clearly the points where the graph crosses the axes.



[2]

20. Simplify $\dfrac{3x^2 - 8x - 16}{(2x + 1)^2 - (x + 3)^2}$.

Answer .... [3]

21. A shop sells bags of groundnuts and walnuts.

A bag of groundnuts contains 25.8 g of protein, 49.2 g of fat and 16.1 g of carbohydrate and costs \$1.60.

A bag of walnuts contains x g of protein, 68.5 g of fat and 4.2 g of carbohydrate and costs \$2.80.

The contents of the bags can be represented by the matrix $\mathbf{A} = \begin{pmatrix} 25.8 & 49.2 & 16.1 \\ x & 68.5 & 4.2 \end{pmatrix}$.

Namita buys 4 bags of groundnuts and 3 bags of walnuts.

Sian buys 5 bags of groundnuts and 2 bags of walnuts.

This information can be represented by the matrix $\mathbf{B} = \begin{pmatrix} 4 & 3 \\ 5 & 2 \end{pmatrix}$.

(a) Find, in terms of x, the matrix $\mathbf{T} = \mathbf{BA}$.

Answer $\mathbf{T} =$ [2]

(b) The nuts bought by Namita contain 5.6 g more protein than Sian's nuts.

Find the value of x.

Answer $x =$ .... [2]

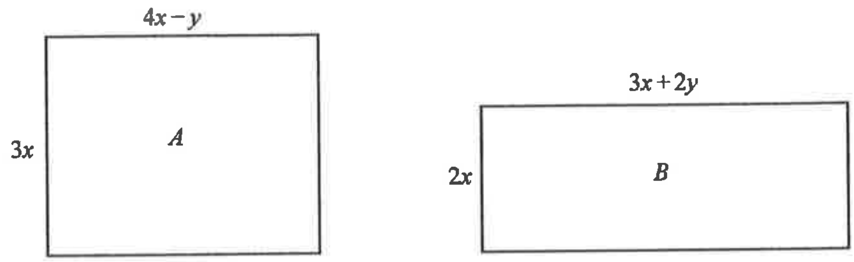
(c) The elements of the matrix $\mathbf{M}$, where $\mathbf{M} = \mathbf{BD}$, represent the total cost, in dollars, of the bags of nuts bought by Namita and Sian.

Write down the matrix $\mathbf{D}$.

Answer $\mathbf{D} =$ [1]

22. The diagram shows two rectangles, A and B.

The dimensions, in centimetres, of the two rectangles are shown.



(a) The two rectangles have equal areas.

Find y in terms of x.

Give your answer in its simplest form.

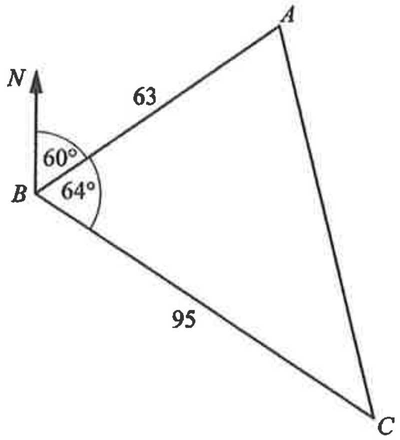
Answer $y =$ .... [2]

(b) The perimeter of rectangle B is 16 cm more than the perimeter of rectangle A.

Find the value of x.

Answer $x =$ .... [3]

23.



A, B and C are three points on horizontal ground.

$BA = 63$ m and the bearing of A from B is 060°.

$BC = 95$ m and angle $ABC = 64°$.

(a) Find the bearing of B from C.

Answer .... [1]

(b) Calculate the distance AC.

Answer $AC =$ .... m [3]

(c) Calculate the bearing of C from A.

Answer .... [3]

24. The intensity, I watts/m^2, of radiation from the Sun is inversely proportional to the square of the distance, D million km, from the Sun.

(a) The average distance of Mars from the Sun is 228 million km.

The average distance of Earth from the Sun is 150 million km.

The intensity of the Sun's radiation at the Earth is approximately 1370 watts/m^2.

When Mars is at its average distance from the Sun, calculate the approximate intensity of the Sun's radiation.

Answer .... watts/m^2 [2]

(b) Explain what happens to the intensity of the Sun's radiation when the distance from the Sun is doubled.

.... [1]

25. The **sum** of the first n terms of a linear sequence is $\frac{1}{2}n(5n + 1)$.

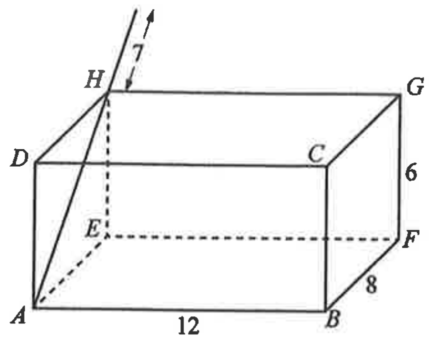
(a) Find the first 3 terms of the sequence.

Answer ...., ..., ... [2]

(b) Find, in terms of n, an expression for the nth term of the sequence.

Answer .... [1]

26.



The diagram shows an open cuboid measuring 12 cm by 8 cm by 6 cm.

A rod is placed inside the cuboid against the face *AEHD*.

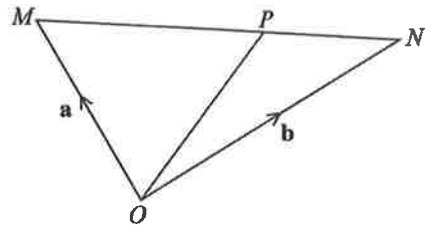
In this position, 7 cm of the rod is outside the cuboid.

The position of the rod is changed so that the length outside the cuboid is as short as possible.

Calculate this shortest length.

Answer .... cm [4]

27.



OMN is a triangle.

P is the point on *MN* such that $MP : PN = 3 : 2$.

$\overrightarrow{OM} = \mathbf{a}$ and $\overrightarrow{ON} = \mathbf{b}$.

(a) Show that $\overrightarrow{OP} = \frac{1}{5}(2\mathbf{a} + 3\mathbf{b})$.

Answer

[2]

(b) *Q* is the point such that $\overrightarrow{MQ} = \frac{1}{5}(\mathbf{a} + 9\mathbf{b})$.

Explain why *O*, *P* and *Q* lie on a straight line.

Answer

....

....

....

.... [3]